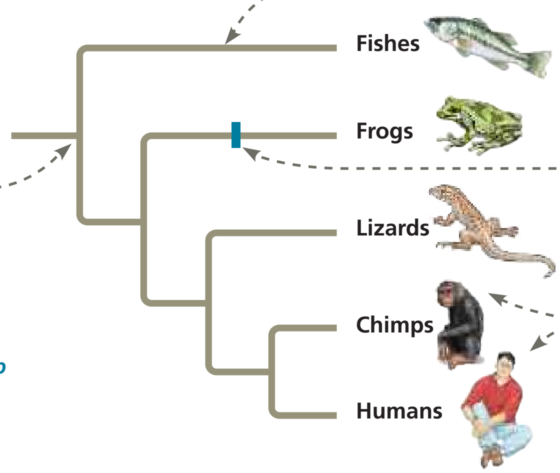
▼ Figure 26.5 VISUALIZING PHYLOGENETIC RELATIONSHIPS

A phylogenetic tree visually represents a hypothesis of how a group of organisms are related. This figure explores how the way a tree is drawn conveys information.

Instructors: Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

Parts of a Phylogenetic Tree

This tree shows how the five groups of organisms at the tips of the branches, called taxa, are related. Each **branch point** represents the common ancestor of the evolutionary lineages diverging from it.



This branch point represents the common ancestor of all the animal groups shown in this tree.

Each horizontal branch represents an **evolutionary lineage.** The length of the branch is arbitrary unless the diagram specifies that branch lengths represent information such as time or amount of genetic change (see Figure 26.13).

Each position along a branch represents an ancestor in the lineage leading to the taxon named at the tip.

Chimps and humans are an example of **sister taxa**, groups of organisms that share a common ancestor that is not shared by any other group. The members of a sister group are each other's closest relatives.

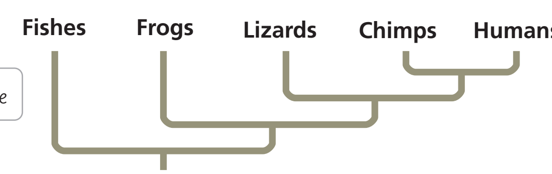
***1.** According to this tree, which group or groups of organisms are most closely related to frogs?*

***2.** Label the part of the diagram that represents the most recent common ancestor of frogs and humans.*

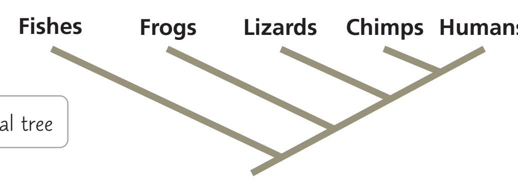
Alternative Forms of Tree Diagrams

These diagrams are referred to as "trees" because they use the visual analogy of branches to represent evolutionary lineages diverging over time. In this text, trees are usually drawn horizontally, as shown above, but the same tree can be drawn vertically or diagonally without changing the relationships it conveys.

Vertical tree



Diagonal tree

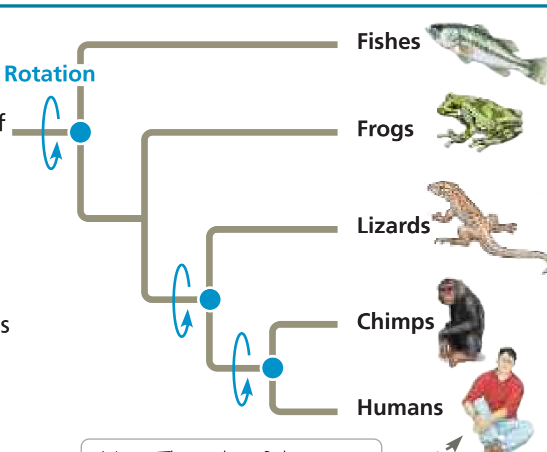


***3.** How many sister taxa are shown in these two trees? Identify them.*

***4.** Redraw the horizontal tree in Figure 26.2 as a vertical tree and a diagonal tree.*

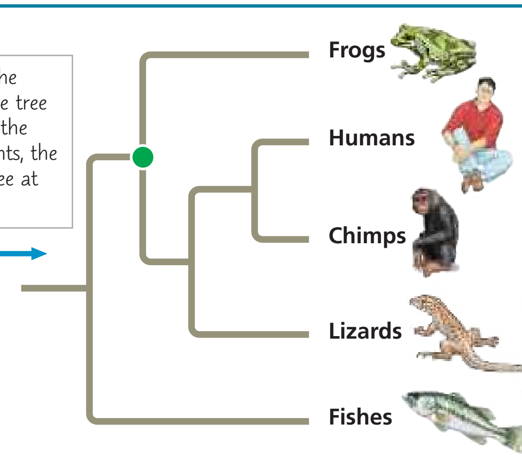
Rotating Around Branch Points

Rotating the branches of a tree around a branch point does not change what they convey about evolutionary relationships. As a result, the order in which taxa appear at the branch tips is not significant. What matters is the branching pattern, which signifies the order in which the lineages have diverged from common ancestors.



Note: The order of the taxa does NOT represent a sequence of evolution "leading to" the last taxon shown (in this tree, humans).

If you rotate the branches of the tree at left around the three blue points, the result is the tree at right.



***5.** Redraw the tree above, rotating around the green branch point. Identify the two closest relatives of humans as shown in each of the three trees. Explain your answer.*